JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON WEDNESDAY 08th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. Consider the relation R on the set {–2, –1, 0, 1, 2} defined by (a, b) ∈ R if and only if 1 + ab > 0. Then among the statements :

I. The number of elements in R is 17

II. R is an equivalence relation

(1) Only I is true (2) Only II is true

(3) Both I and II are true (4) Neither I nor II is true

Sol. Reflexive If $(a, a) \in R$

$\Rightarrow 1 + a^2 > 0$

$\Rightarrow$ It is reflexive relation

Symmetric If $1 + ab > 0 \Rightarrow 1 + ba > 0$

i.e. If $(a, b) \in R$ and $(b, a) \in R$

Therefore it is symmetric relation

Transitive $\because (-2, 0) \in R$ & $(0, 2) \in R$

But $(-2, 2) \notin R$

Hence it is not transitive

$\Rightarrow$ It is not an equivalence relation.

Check Number of elements in R

$\because 1 + ab > 0$

a & b each can be filled by 5 ways

$\therefore$ total elements = 25 But out of these

(–2, –1), (–2, 2), (–1, 2), (1, –2), (2, –2), (2, –1)

(–1, 1) & (1, –1) are not in relation.

2. The number of values of $z \in \mathbb{C}$, satisfying the equations $|z - (4 + 8i)| = \sqrt{10}$ and $|z - (3 + 5i)| + |z - (5 + 11i)| = 4\sqrt{5}$, is :

(1) 0 (2) 2

(3) 1 (4) 4

Sol. $|z - (3 + 5i)| + |z - (5 + 11i)| = 4\sqrt{5}$

So, $S_1(3, 5)$ & $S_2(5, 11)$ are two foci of ellipse,

where $2ae = S_1S_2 = 2\sqrt{10}$ & $2a = 4\sqrt{5}$

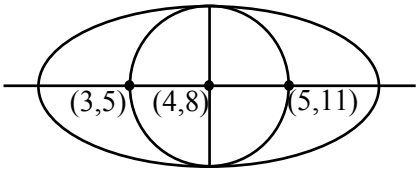
$\Rightarrow a = 2\sqrt{5}$, $e = \frac{1}{\sqrt{2}}$ & $b^2 = a^2(1 - e^2)$ gives $b^2 = 10$

& center of an ellipse is mid point of both focus

$\therefore$ center (4, 8)

Equation of circle is $|Z - (4 + 8i)| = \sqrt{10}$

$\Rightarrow$ Ellipse & given circle are concentric & radius of given circle is equal to length of minor axis of given ellipse



$\Rightarrow$ Two points which are common to both.

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3. If the system of linear equations :

$x + y + z = 6$,

$x + 2y + 5z = 10$,

$2x + 3y + \lambda z = \mu$.

has infinitely many solutions, then the value of $\lambda + \mu$ equals.

(1) 12 (2) 16

(3) 22 (4) 28

Sol. $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \\ 2 & 3 & \lambda \end{vmatrix} = \lambda - 6$

$D_1 = \begin{vmatrix} 6 & 1 & 1 \\ 10 & 2 & 5 \\ \mu & 3 & \lambda \end{vmatrix} = 2\lambda + 3\mu - 60$

$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 10 & 5 \\ 2 & \mu & \lambda \end{vmatrix} = 4(\lambda - \mu + 10)$

$D_3 = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 10 \\ 2 & 3 & \mu \end{vmatrix} = \mu - 16$

For infinite many solution

$D = D_1 = D_2 = D_3 = 0$

$\Rightarrow \lambda = 6$ & $\mu = 16$

$\Rightarrow \lambda + \mu = 22$

4. Let $A = \begin{bmatrix} \alpha & 1 & 2 \\ 2 & 3 & 0 \\ 0 & 4 & 5 \end{bmatrix}$ and

$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -5\alpha & 0 \\ 0 & 4\alpha & -2\alpha \end{bmatrix}$ + adj (A). If det(B) = 66,

then det(adj (A)) equals :

(1) 289 (2) 361

(3) 441 (4) 529

Sol. $A = \begin{bmatrix} \alpha & 1 & 2 \\ 2 & 3 & 0 \\ 0 & 4 & 5 \end{bmatrix}$ & $|A| = 5\alpha + 6$

Now adj $A = \begin{bmatrix} 15 & 3 & -6 \\ -10 & 5\alpha & 4 \\ 8 & -4\alpha & 3\alpha - 2 \end{bmatrix}$

$\therefore B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -5\alpha & 0 \\ 0 & 4\alpha & -2\alpha \end{bmatrix}$ + adj(A)

$B = \begin{bmatrix} 16 & 3 & -6 \\ -10 & 0 & 4 \\ 8 & 0 & \alpha - 2 \end{bmatrix}$

& $|B| = 6(5\alpha + 6) = 66$ (given in question)

$\Rightarrow \alpha = 1$

$\Rightarrow |A| = 21$

$\therefore$ det(adj A) = $|A|^{(n-1)^2}$

Here n = 3

$= 21^2$

$= 441$

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5. Let $\alpha = 3 + 4 + 8 + 9 + 13 + 14 + \ldots..$ upto 40 terms. If $(\tan\beta)^{\frac{\alpha}{1020}}$ is a root of the equation $x^2 + x - 2 = 0$, $\beta \in \left(0, \frac{\pi}{2}\right)$, then $\sin^2\beta + 3\cos^2\beta$ is equal to :

(1) 2 (2) $\frac{7}{4}$

(3) $\frac{5}{2}$ (4) $\frac{3}{2}$

Sol. a = (3 + 8 + 13 … upto 20 terms) + (4 + 9 + 14 + … upto 20 terms)

$$= \frac{20}{2}[6+19\times5] + \frac{20}{2}(8+19\times5)$$

$$= 2040$$

$$(\tan\beta)^{\frac{2040}{1020}} = \tan^2\beta$$

$x^2 + x - 2 = 0 \Rightarrow 1, -2$

$\Rightarrow \tan^2\beta = 1$; $\sin^2\beta = \frac{1}{2}$ & $\cos^2\beta = \frac{1}{2}$

$$\sin^2\beta + 3\cos^2\beta = \frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$$

6. A candidate has to go to the examination centre to appear in an examination. The candidate uses only one means of transportation for the entire distance out of bus, scooter and car. The probabilities of the candidate going by bus, scooter and car, respectively, are $\frac{2}{5}$, $\frac{1}{5}$ and $\frac{2}{5}$. The probabilities that the candidate reaches late at the examination centre are $\frac{1}{5}$, $\frac{1}{3}$ and $\frac{1}{4}$ if the candidate uses bus, scooter and car, respectively. Given that the candidate reached late at the examination centre, the probability that the candidate travelled by bus is :

(1) $\frac{11}{37}$ (2) $\frac{12}{37}$

(3) $\frac{13}{37}$ (4) $\frac{14}{37}$

Sol.

$P(\text{Bus}) = P(B) = \frac{2}{5}$	$P\left(\frac{\text{late}}{B}\right) = \frac{1}{5}$
$P(\text{Scooter}) = P(S) = \frac{1}{5}$	$P\left(\frac{\text{late}}{S}\right) = \frac{1}{3}$
$P(\text{Car}) = P(C) = \frac{2}{5}$	$P\left(\frac{\text{late}}{C}\right) = \frac{1}{4}$

$$P\left(\frac{B}{\text{late}}\right) = \frac{\frac{2}{5}\times\frac{1}{5}}{\frac{2}{5}\times\frac{1}{5}\times\frac{1}{5}\times\frac{1}{3}\times\frac{2}{5}\times\frac{1}{4}} = \frac{12}{37}$$

7. A set of four observations has mean 1 and variance 13. Another set of six observations has mean 2 and variance 1. Then, the variance of all these 10 observations is equal to :

(1) 5.96 (2) 6.14

(3) 6.04 (4) 6.24

Sol. Given $\bar{x} = 1, \sigma_1^2 = 13$

$\bar{y} = 2, \sigma_2^2 = 1$

Combined variance = $\frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1+n_2)^2}(\bar{x} - \bar{y})^2$

$$= \frac{4\times13 + 6\times1}{10} + \frac{4\times6}{10^2}(2-1)^2$$

$$\frac{58}{10} + \frac{24}{100} = \frac{580+24}{100} = \frac{604}{100} = 6.04$$

8. If

$$26\left(\frac{2^3}{3}\left({}^{13}C_2\right) + \frac{2^5}{5}\left({}^{12}C_4\right) + \frac{2^7}{7}\left({}^{12}C_6\right) + \ldots. + \frac{2^{13}}{13}\left({}^{12}C_{12}\right)\right)$$

$= 3^{13} - \alpha$, then α is equal to :

(1) 45 (2) 48

(3) 51 (4) 54

Sol. $(1+x)^{12} = {}^{12}C_0 + {}^{12}C_1 x + {}^{12}C_2 x^2 + \ldots\ldots + {}^{12}C_{12} x^{12}$

integrating both sides

$$\left(\frac{(1+x)^{13}}{13}\right)_{-2}^{2} = \left({}^{12}C_0 x + {}^{12}C_1\frac{x^2}{2} + {}^{12}C_2\frac{x^3}{3} + \ldots + {}^{12}C_{12}\frac{x^{13}}{13}\right)_{-2}^{2}$$

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$$\frac{3^{13}+1}{13}=\left({}^{12}C_0 2+{}^{12}C_1\frac{2^2}{2}+{}^{12}C_2\frac{2^3}{3}+\ldots+{}^{12}C_{12}\frac{2^{13}}{13}\right)$$

$$-\left(-{}^{12}C_0.2+{}^{12}C_1\frac{2^2}{2}-{}^{12}C_2.\frac{2^3}{3}+\ldots+{}^{12}C_{12}.\frac{2^{13}}{13}\right)$$

$$=2\left(2.{}^{12}C_0+{}^{12}C_2.\frac{2^3}{3}+{}^{12}C_4.\frac{2^5}{5}+\ldots+{}^{12}C_{12}.\frac{2^{13}}{13}\right)$$

$$3^{13}+1=26\left(2+{}^{12}C_{12}.\frac{2^3}{3}+{}^{12}C_4.\frac{2^5}{5}+\ldots+{}^{12}C_{12}.\frac{2^{13}}{13}\right)$$

$$3^{13}-51=26\left({}^{12}C_2.\frac{2^3}{3}+{}^{12}C_4.\frac{2^5}{5}+\ldots+{}^{12}C_{12}.\frac{2^{13}}{13}\right)$$

$\therefore \alpha = 51$

9. A person has three different bags and four different books. The number of ways, in which he can put these books in the bags so that no bag is empty, is

(1) 18 (2) 36

(3) 39 (4) 72

Sol. B_1 B_2 B_3

1 1 2

No. of ways $= \frac{4!}{2!2!}\times 3! = 6\times 6 = 36$

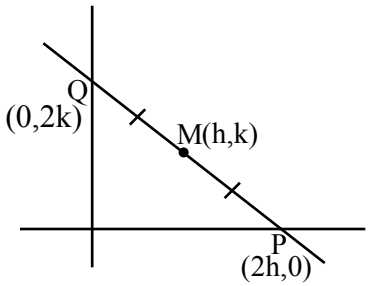
10. If a straight line drawn through the point of intersection of the lines $4x + 3y - 1 = 0$ and $3x + 4y - 1 = 0$, meet the co-ordinate axes at the points P and Q, then the locus of the mid point of PQ is:

(1) $x + y - 7 = 0$ (2) $x + y - 14xy = 0$

(3) $2x + y + 14xy = 0$ (4) $x + 2y - 14xy = 0$

Sol. Point of intersection of the lines

$3x + 4y = 1$ and $4x + 3y = 1$ is $\left(\frac{1}{7},\frac{1}{7}\right)$



Let the line be $\frac{x}{2h}+\frac{y}{2k}=1$

Satisfy $\left(\frac{1}{7},\frac{1}{7}\right)$

$$\frac{1}{14h}+\frac{1}{14k}=1$$

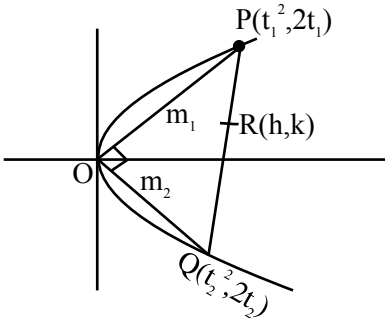
$$\frac{1}{x}+\frac{1}{y}=14$$

11. Let O be the vertex of the parabola $y^2 = 4x$ and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point of the line segment PQ is a conic C, then the length of its latus rectum is :

(1) 1 (2) 2

(3) 4 (4) 8

Sol.



$t_1t_2 = -4$

$2h = t_1^2 + t_2^2$ $\qquad 2k = 2(t_1 + t_2)$

$2h = k^2 - 2t_1t_2$

$2h = k^2 + 8$

$k^2 = 2h - 8$

$= 2((h - 4)$

L.R. = 2

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12. Let $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, where inverse trigonometric functions take only the principal values.

Given below are two statements:

Statement I : $\cos(\alpha+\beta) > 0$

Statement II : $\cos(\alpha) < 0$

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol. $\frac{1}{2} < \frac{6}{11} < \frac{1}{\sqrt{2}}$

$\sin^{-1}\left(\frac{1}{2}\right) < \sin^{-1}\left(\frac{6}{11}\right) < \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

$\frac{\pi}{6} < 3\sin^{-1}\left(\frac{6}{11}\right) < \frac{\pi}{4}$

$\frac{\pi}{2} < \alpha < \frac{3\pi}{4} \quad \therefore \cos\alpha < 0$

$0 < \frac{4}{9} < \frac{1}{2}$

$\frac{\pi}{3} < \cos^{-1}\left(\frac{4}{9}\right) < \frac{\pi}{2}$

$\pi < 3\cos^{-1}\left(\frac{4}{9}\right) < \frac{3\pi}{2}$

$\pi < \beta < \frac{3\pi}{2}$

Now $\frac{3\pi}{2} < \alpha+\beta < \frac{9\pi}{4}$

$\therefore \cos(\alpha+\beta) > 0$

13. For the function $f(x) = e^{\sin|x|} - |x|$, $x \in \mathbf{R}$, consider the following statements:

Statement I : f is differentiable for all $x \in \mathbf{R}$.

Statement II : f is increasing in $\left(-\pi, -\frac{\pi}{2}\right)$.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol. $f(x) = e^{|\sin x|} - |x|$ is

non-differentiable at $x = \pi$

and for $x \in \left(-\pi, \frac{-\pi}{2}\right)$

$f(x) = e^{-\sin x} + x$

$f'(x) = -e^{-\sin x}\cos x + 1$

for $x \in \left(-\pi, \frac{-\pi}{2}\right)$ $\quad \cos x < 0$

$\therefore f'(x) > 0$

$\therefore$ f(x) is increasing

14. Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a}\times\vec{b}) + 3(\vec{b}\times\vec{c}) = \vec{0}$. If $\vec{a}.\vec{c} = 15$, then $\vec{c}\cdot(\hat{i}+\hat{j}-3\hat{k})$ is equal to :

(1) –6 (2) –5

(3) –4 (4) –3

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Sol. $2(\vec{a}\times\vec{b})-3(\vec{c}\times\vec{b})=\vec{0}$

$(2\vec{a}-3\vec{c})\times\vec{b}=\vec{0}$

$\vec{b}\parallel(2\vec{a}-3\vec{c})$

$2\vec{a}-3\vec{c}=\lambda\vec{b}$

$\vec{c}=\dfrac{2\vec{a}-\lambda\vec{b}}{3}$ ...(1)

$\vec{a}.\vec{c}=15$

$\left(\dfrac{2\vec{a}-\lambda\vec{b}}{3}\right).\vec{a}=15$

$\dfrac{2|a|^2-\lambda\vec{b}.\vec{a}}{3}=15$

$2(26)-\lambda(40-2-3)=45$

$\lambda=\dfrac{1}{5}$

$\Rightarrow \vec{c}.(\hat{i}+\hat{j}-3\hat{k})=\dfrac{\left(2(4\hat{i}-\hat{j}+3k)-\frac{1}{5}(10\hat{i}+2\hat{j}-k).(\hat{i}+\hat{j}-3\hat{k})\right)}{3}$

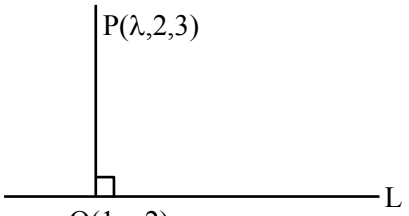
$=\dfrac{2(4-1-9)-\frac{1}{5}(10+2+3)}{3}=-5$

15. Let the foot of perpendicular from the point $(\lambda, 2, 3)$ on the line $\dfrac{x-4}{1}=\dfrac{y-9}{2}=\dfrac{z-5}{1}$ be the point $(1, \mu, 2)$. Then the distance between the lines $\dfrac{x-1}{2}=\dfrac{y-2}{3}=\dfrac{z+4}{6}$ and $\dfrac{x-\lambda}{2}=\dfrac{y-\mu}{3}=\dfrac{z+5}{6}$ is equal to

(1) $\dfrac{12}{7}$ (2) $\dfrac{\sqrt{145}}{7}$

(3) $\dfrac{\sqrt{146}}{7}$ (4) $\dfrac{\sqrt{143}}{7}$

Sol.



$\dfrac{x-4}{1}=\dfrac{y-9}{2}=\dfrac{z-5}{1}$

Q(1, μ, 2) satisfies the line L

$\therefore \dfrac{1-4}{1}=\dfrac{\mu-9}{2}=\dfrac{2-5}{1}$

$\Rightarrow -3=\dfrac{\mu-9}{2}=-3 \qquad \Rightarrow \boxed{\mu=3}$

D.R. of PQ : $\lambda-1, 2-\mu, 1$

Since PQ $\perp$ L $\Rightarrow (\lambda-1).1+2(2-\mu)+1.1=0$

$\Rightarrow \boxed{\lambda-2\mu+4=0} \qquad \therefore \boxed{\lambda=2}$

Now lines : $\dfrac{x-1}{2}=\dfrac{y-2}{3}=\dfrac{z+4}{6}$ &

$\dfrac{x-2}{2}=\dfrac{y-3}{3}=\dfrac{z+5}{6}$

i.e. $\vec{r}=\left(\hat{i}+2\hat{j}-4\hat{k}\right)+t_1\left(2\hat{i}+3\hat{j}+6\hat{k}\right)$

& $\vec{r}=\left(2\hat{i}+3\hat{j}-5\hat{k}\right)+t_2\left(2\hat{i}+3\hat{j}+6\hat{k}\right)$

Distance

$=\dfrac{\left|\left(\left(\hat{i}+2\hat{j}-4\hat{k}\right)-\left(2\hat{i}+3\hat{j}-5\hat{k}\right)\right)\times\left(2\hat{i}+3\hat{j}+6\hat{k}\right)\right|}{\left|2\hat{i}+3\hat{j}+6\hat{k}\right|}$

$=\dfrac{\left|-9\hat{i}+8\hat{j}-\hat{k}\right|}{7}=\dfrac{\sqrt{146}}{7}$

16. The value of the integral

$\displaystyle\int_0^2 \frac{\sqrt{x\left(x^2+x+1\right)}}{\left(\sqrt{x+1}\right)\left(\sqrt{x^4+x^2+1}\right)}\,dx$ is equal to :

(1) $\dfrac{1}{3}\log_e\left(3-2\sqrt{2}\right)$

(2) $\dfrac{2}{3}\log_e\left(4+\sqrt{2}\right)$

(3) $\dfrac{2}{3}\log_e\left(3+2\sqrt{2}\right)$

(4) $\dfrac{1}{3}\log_e\left(1+6\sqrt{2}\right)$

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Sol. $I = \int_0^2 \sqrt{\frac{x(x^2+x+1)}{(x+1)(x^2-x+1)(x^2+x+1)}}dx$

$= \int_0^2 \sqrt{\frac{x}{x^3+1}}dx$

$(x^{3/2} = t \Rightarrow \frac{3}{2}x^{1/2}dx = dt)$

$= \frac{2}{3}\int_0^{2^{3/2}} \frac{dt}{\sqrt{t^2+1}}$

$= \frac{2}{3}\left(\ell n(t+\sqrt{t^2+1})\right)_0^{2^{3/2}}$

$= \frac{2}{3}\ell n\left(2^{3/2}+3\right)$

17. Let y = y(x) be the solution of the differential equation

$x\sqrt{1-x^2}\,dy + \left(y\sqrt{1-x^2} - x\cos^{-1}x\right)dx = 0,$

$x \in (0, 1)$, $\lim\limits_{x\to 1^-} y(x) = 1$. Then $y\left(\frac{1}{2}\right)$ equals :

(1) $3 - \frac{\pi}{\sqrt{3}}$ (2) $4 - \sqrt{3}\pi$

(3) $4 - \frac{2\pi}{\sqrt{3}}$ (4) $3 - \frac{\pi}{2\sqrt{3}}$

Sol. $x\sqrt{1-x^2}dy + \left(y\sqrt{1-x^2} - x\cos^{-1}x\right)dx = 0$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{\cos^{-1}x}{\sqrt{1-x^2}}$

it is L.D.E.

$\therefore$ I.F. $= e^{\int \frac{1}{x}dx} = e^{\ell n\, x} = x$

$\therefore$ solution is $y.x = \int \frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx + C$

$y.x = I_1 + C$ ....(1)

$I_1 = \int \frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx$

Let $\cos^{-1}x = t \Rightarrow x = \cos t$

$-\frac{1}{\sqrt{1-x^2}}dx = dt$

$I_1 = \int -t.\cot\, dt$

$= -(t\sin t + \cos t)$

$I_1 = -\left(\sqrt{1-x^2}.\cos^{-1}x + x\right)$ ....(2)

$\therefore\ y.x = -\left(\sqrt{1-x^2}.\cos^{-1}x + x\right) + C$

$\because \lim\limits_{x\to 1^-} y(x) = 1$

$\therefore\ 1 = -(0+1) + C$

$\Rightarrow C = 2$

$\therefore\ y.x = -\sqrt{1-x^2}.\cos^{-1}x - x + 2$

$\left(\text{put } x = \frac{1}{2}\right)$

$y\left(\frac{1}{2}\right)\times\frac{1}{2} = -\frac{\sqrt{3}}{2}\times\frac{\pi}{3} - \frac{1}{2} + 2$

$\Rightarrow y\left(\frac{1}{2}\right) = 3 - \frac{\pi}{\sqrt{3}}$

18. Let $f : (1, \infty) \to \mathbf{R}$ be function defined as $f(x) = \frac{x-1}{x+1}$. Let $f^{i+1}(x) = f\left(f^i(x)\right)$, $i = 1, 2, \ldots, 25$, where $f^1(x) = (x)$. If $g(x) + f^{26}(x) = 0$, $x \in (1, \infty)$, then the area of the region bounded by the curves $y = g(x)$, $2y = 2x - 3$, $y = 0$ and $x = 4$ is:

(1) $\frac{1}{8} + \log_e 2$ (2) $\frac{1}{4} + \log_e 2$

(3) $\frac{5}{6} + 3\log_e 2$ (4) $\frac{5}{6} + \log_e 2$

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Sol. $f^2(x) = f(f(x)) = \dfrac{\dfrac{x-1}{x+1}-1}{\dfrac{x-1}{x+1}+1} = \dfrac{-1}{x}$

$f^3(x) = f(f^2(x)) = f\left(\dfrac{-1}{x}\right) = \dfrac{1+x}{1-x}$

$f^4(x) = f(f^3(x)) = f\left(f\left(\dfrac{-1}{x}\right)\right) = x$

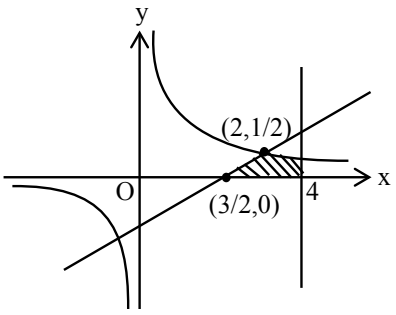
$f^5(x) = f(f^4(x)) = f(x)$

$f^6(x) = f(f^5(x)) = f(f(x)) = \dfrac{-1}{x}$

.
.
.

$f^{26}(x) = \dfrac{-1}{x}$

$g(x) = \dfrac{1}{x}$



$x - 3/2 = 1/x \Rightarrow 2x^2 - 3x - 2 = 0$

$\Rightarrow x = -1/2, 2$

Req. area $= \int_2^4 \dfrac{1}{x}dx + \dfrac{1}{2}\times\dfrac{1}{2}\times\dfrac{1}{2}$

$= \dfrac{1}{8} + (\ell nx)_2^4 = \dfrac{1}{8} + \ell n2$

19. Let $f(x) = \begin{cases} \dfrac{1}{3}, & x \le \pi/2 \\ \dfrac{b(1-\sin x)}{(\pi-2x)^2}, & x > \pi/2 \end{cases}$, If f is continuous at $x = \pi/2$, then the value of $\int_0^{3b-6} \left|x^2+2x-3\right| dx$ is :

(1) 5 (2) 2

(3) 3 (4) 4

Sol. $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{b(1-\cos(\frac{\pi}{2}-x))}{4\left(\frac{\pi}{2}-x\right)^2} = \dfrac{1}{3} \Rightarrow b = \dfrac{8}{3}$

$\int_0^2 \left|x^2+2x-3\right| dx = \int_0^2 \left|(x+3)(x-1)\right| dx$

put $x - 1 = t$

$\int_{-1}^1 \left|(t+4)t\right| dt = \int_{-1}^0 (-t^2-4t)dt + \int_0^1 (t^2+4t)dt$

$\left(\dfrac{-t^3}{3} - 2t^2\right)_{-1}^0 + \left(\dfrac{t^3}{3} + 2t^2\right)_0^1 = 4$

20. Let $\dfrac{x^2}{f\left(a^2+7a+3\right)} + \dfrac{y^2}{f\left(3a+15\right)} = 1$ represent an ellipse with major axis along y-axis, where f is a strictly decreasing positive function on **R**. If the set of all possible values of a is **R** $-[\alpha, \beta]$, then $\alpha^2 + \beta^2$ is equal to :

(1) 28 (2) 40

(3) 61 (4) 4

Sol. Given ellipse is vertical

$\therefore f(3a + 15) > f(a^2 + 7a + 3)$

$\because f(x)$ is decreasing $\forall\ x \in R$

$\Rightarrow 3a + 15 < a^2 + 7a + 3$

$\Rightarrow a^2 + 4a - 12 > 0$

$\Rightarrow (a + 2)^2 > 16$

$\Rightarrow |(a + 2)| > 4$

$\Rightarrow a > 2$ or $a < -6$

$\Rightarrow a \in R - [-6, 2]$

$\therefore \alpha = -6, \beta = 2$

$\therefore \alpha^2 + \beta^2 = 36 + 4 = 40$

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SECTION-B

21. The sum of squares of all real solution of the equation

$\log_{(x+1)}(2x^2 + 5x + 3) = 4 - \log_{(2x+3)}(x^2 + 2x + 1)$ is equal to _____.

Sol. $\log_{(x+1)}(2x+3)(x+1) + \log_{(2x+3)}(x+1)^2 = 4$

$1 + \log_{(x+1)}(2x+3) + 2\log_{(2x+3)}(x+1) = 4$

$\log_{x+1}(2x+3) = t \qquad t + \frac{2}{t} = 3$

$t^2 - 3t + 2 = 0 \Rightarrow t = 1, 2$

for $t = 1 \Rightarrow \log_{(x+1)}(2x+3) = 1 \Rightarrow 2x + 3 = x + 1$

$\Rightarrow x = -2$ (rejected)

for $t = 2 \Rightarrow 2x + 3 = x^2 + 2x + 1 \Rightarrow x^2 = 2$

$\Rightarrow x = \pm\sqrt{2}$

rejecting $x = -\sqrt{2}$, we get $x = \sqrt{2}$

$\therefore$ sum of squares of all the roots = 2

22. If $\int_{\pi/6}^{\pi/4}\left(\cot\left(x-\frac{\pi}{3}\right)\cot\left(x+\frac{\pi}{3}\right)+1\right)dx = \alpha \log_e\left(\sqrt{3}-1\right)$, then $9\alpha^2$ is equal to _____.

Sol. $\int_{\pi/6}^{\pi/4}\left((\cot(x-\frac{\pi}{3})\cot(x+\frac{\pi}{3})+1\right)dx$

using $\cot(A-B) = \frac{\cot B\cot A + 1}{\cot B - \cot A}$

$\Rightarrow \int_{\pi/6}^{\pi/4} -\frac{1}{\sqrt{3}}\left[\cot(x-\frac{\pi}{3}) - \cot(x+\frac{\pi}{3})\right]dx$

$\Rightarrow \frac{-1}{\sqrt{3}}\left[\ell n\left|\frac{\sin(x-\frac{\pi}{3})}{\sin(x+\frac{\pi}{3})}\right|\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$

$= \frac{-1}{\sqrt{3}}\left[\ell n\left|\left(\frac{\sin 15^\circ}{\sin 105^\circ}\right)\right| - \ell n\left|\frac{\sin 30^\circ}{\sin 90^\circ}\right|\right]$

$= \frac{-1}{\sqrt{3}}\left[\ell n(\tan 15^\circ) - \ell n\left(\frac{1}{2}\right)\right.$

$= \frac{-1}{\sqrt{3}}\left[\ell n\left(2-\sqrt{3}\right) + \ell n 2\right]$

$\Rightarrow \frac{-1}{\sqrt{3}}\ell n\left(4-2\sqrt{3}\right) = \frac{-2}{\sqrt{3}}\ell n\left(\sqrt{3}-1\right)$

$\therefore \alpha = \frac{-2}{\sqrt{3}} \therefore 9\alpha^2 = 12$

23. Let a line L_1 pass through the origin and be perpendicular to the lines

$L_2 : \vec{r} = (3+t)\hat{i} + (2t-1)\hat{j} + (2t+4)\hat{k}$ and

$L_3 : \vec{r} = (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}$, t, $s \in \mathbf{R}$. If (a, b, c), $a \in \mathbf{Z}$, is the point on L_3 at a distance of $\sqrt{17}$ from the point of intersection of L_1 and L_2, then $(a+b+c)^2$ is equal to _____.

Sol. $L_2 : \frac{x-3}{1} = \frac{y+1}{2} = \frac{z-4}{2} = t$

$L_3 : \frac{x-3}{2} = \frac{y-3}{2} = \frac{z-2}{1} = s$

vector $\perp^{ar}$ to L_2 & L_3

$\Rightarrow \begin{vmatrix} i & j & k \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} = \hat{i}(-2) - \hat{j}(-3) + \hat{k}(-2)$

$\Rightarrow -2\hat{i} + 3\hat{j} - 2\hat{k}$

$\Rightarrow -\left(2\hat{i} - 3\hat{j} + 2\hat{k}\right)$

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Now, $L_1 : \frac{x}{2} = \frac{y}{-3} = \frac{z}{2} = \ell$

intersection of L_1 & L_2

$\Rightarrow t + 3 = 2\ell,\ 2t - 1 = -3\ell,\ 2t + 4 = 2\ell$

$t + 3 = 2t + 4$

$t = -1$

$\Rightarrow P \equiv (2, -3, 2)$

A point on L_3 is $Q(2s + 3, 2s + 3, s + 2)$

$\Rightarrow PQ^2 = (2s + 1)^2 + (2s + 6)^2 + s^2 = 17$

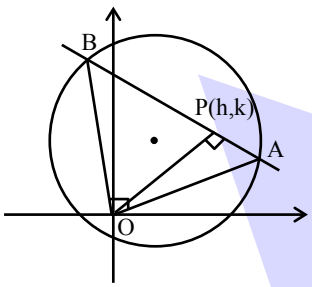
$\Rightarrow 9s^2 + 28s + 20 = 0$

$\Rightarrow (9s + 10)(s + 2) = 0 \Rightarrow s = -2$

$\therefore Q \equiv (-1, -1, 0) \equiv (a, b, c) \Rightarrow (a + b + c)^2 = 4$

24. Consider the circle $C : x^2 + y^2 - 6x - 8y - 11 = 0$. Let a variable chord AB of the circle C subtend a right angle at the origin. If the locus of the foot of the perpendicular drawn from the origin on the chord AB is the circle $x^2 + y^2 - \alpha x - \beta y - \gamma = 0$, then $\alpha + \beta + 2\gamma$ is equal to _____.

Sol.



$C : x^2 + y^2 - 6x - 8y - 11 = 0$

E^{qn} of chord AB

$y - k = -\frac{h}{k}(x - h)$

$hx + ky = h^2 + k^2$ ....(2)

Homogenising the eq^n of the circle

$x^2 + y^2 - 6x\left(\frac{hx + ky}{h^2 + k^2}\right) - 8y\left(\frac{hx + ky}{h^2 + k^2}\right) - 11\left(\frac{hx + ky}{h^2 + k^2}\right)^2 = 0$

coeff. of x^2 + coeff of $y^2 = 0$

$1 - \frac{6h}{h^2 + k^2} - 11.\frac{h^2}{(h^2 + k^2)^2} + 1 - \frac{8k}{h^2 + k^2} - 11.\frac{k^2}{(h^2 + k^2)^2} = 0$

$2(h^2 + k^2) - 6h - 8k - 11 = 0$

$x^2 + y^2 - 3x - 4y - \frac{11}{2} = 0$

$\therefore \alpha + \beta + 2\gamma = 3 + 4 + 11 = 18$

25. Let f be a polynomial function such that

$\log_2(f(x)) = \left(\log_2\left(2 + \frac{2}{3} + \frac{2}{9} + ...\infty\right)\right)$.

$\log_3\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$, $x > 0$ and $f(6) = 37$. Then

$\sum_{n=1}^{10} f(n)$ is equal to _____.

Sol. $\log_2 f(x) = \log_2\left(\frac{2}{1 - \frac{1}{3}}\right).\log_3\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$

$\log_2 f(x) = \log_2 3.\log_3\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$

$\Rightarrow f(x) = 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}$

$\Rightarrow f(x)\ f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$

$\Rightarrow f(x) = 1 \pm x^n$ $\Rightarrow f(6) = 37$

$\Rightarrow 1 \pm 6^n = 37$ $\Rightarrow 6^n = 36$

$\Rightarrow n = 2$

$\therefore f(x) = 1 + x^2$

$\sum_{n=1}^{10}(1 + n^2) = 10 + \frac{10.11.21}{6} = 395$

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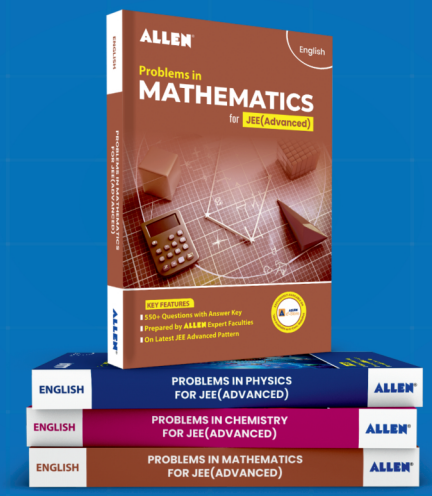


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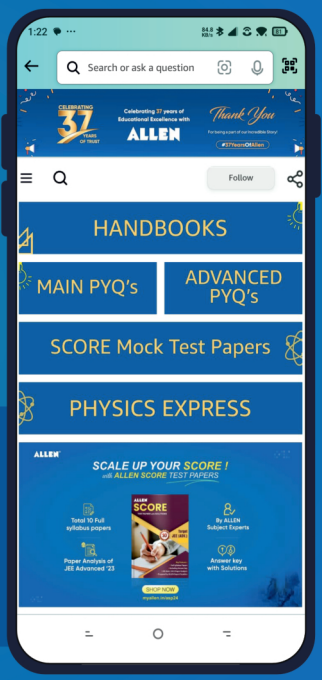
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